Breaking prolonged sitting reduces postprandial glycemia in healthy, normal-weight adults: a randomized crossover trial.

BACKGROUND: Sedentary behavior is a risk factor for cardiometabolic disease. Regularly interrupting sedentary behavior with activity breaks may lower this risk. OBJECTIVE: We compared the effects of prolonged sitting, continuous physical activity combined with prolonged sitting, and regular activity breaks on postprandial metabolism. DESIGN: Seventy adults participated in a randomized crossover study. The prolonged sitting intervention involved sitting for 9 h, the physical activity intervention involved walking for 30 min and then sitting, and the regular-activity-break intervention involved walking for 1 min 40 s every 30 min. Participants consumed a meal-replacement beverage at 60, 240, and 420 min. RESULTS: The plasma incremental area under the curve (iAUC) for insulin differed between interventions (overall P < 0.001). Regular activity breaks lowered values by 866.7 IU · L(-1) · 9 h(-1) (95% CI: 506.0, 1227.5 IU · L(-1) · 9 h(-1); P < 0.001) when compared with prolonged sitting and by 542.0 IU · L(-1) · 9 h(-1) (95% CI: 179.9, 904.2 IU · L(-1) · 9 h(-1); P = 0.003) when compared with physical activity. Plasma glucose iAUC also differed between interventions (overall P < 0.001). Regular activity breaks lowered values by 18.9 mmol · L(-1) · 9 h(-1) (95% CI: 10.0, 28.0 mmol · L(-1) · 9 h(-1); P < 0.001) when compared with prolonged sitting and by 17.4 mmol · L(-1) · 9 h(-1) (95% CI: 8.4, 26.3 mmol · L(-1) · 9 h(-1); P < 0.001) when compared with physical activity. Plasma triglyceride iAUC differed between interventions (overall P = 0.023). Physical activity lowered values by 6.3 mmol · L(-1) · 9 h(-1) (95% CI: 1.8, 10.7 mmol · L(-1) · 9 h(-1); P = 0.006) when compared with regular activity breaks. CONCLUSION: Regular activity breaks were more effective than continuous physical activity at decreasing postprandial glycemia and insulinemia in healthy, normal-weight adults. This trial was registered with the Australian New Zealand Clinical Trials registry as ACTRN12610000953033.